

CHILDREN'S ABILITY TO MAKE TRANSITIVE INFERENCES: THE IMPORTANCE OF PREMISE INTEGRATION AND STRUCTURAL COMPLEXITY

**Glenda Andrews
Graeme S. Halford**
University of Queensland

Four experiments investigated 4- to 6-year-olds' transitive inferences. In Experiments 1-3, there was a nonmapping condition in which inferences were made either about stacked blocks, or about sticks ordered left to right. In the mapping condition, inferences were made by mapping either from blocks to sticks, or the reverse. In Experiments 2-4, relational complexity was manipulated by requiring either 1 or 2 premise relations to be processed in a single decision. Mapping was harder than nonmapping, but relational complexity was the main source of variance, with 2 relations being harder in both mapping and nonmapping conditions. The percentage of participants integrating 2 relations in a single decision was estimated at 20% at age 4, 53% at age 5 and 57% at age 6, suggesting gradual development of transitive inference ability. Results suggest that relational complexity has a strong effect on transitive inference in 4- to 6-year-old children.

If adults are told that John is taller than Peter, and Peter is taller than Mary, most can infer that John is taller than Mary. More generally, for any transitive relation R , if $A R B$ and $B R C$, then $A R C$. Such reasoning is known as transitive inference. Transitivity of an asymmetric binary relation defines an ordered set, therefore it is fundamental to understanding many basic quantitative concepts. Although transitivity seems simple to adults, it causes difficulty for children under

We would like to thank Christine Boag for her excellent work on Experiment 4, and we would like to acknowledge two anonymous reviewers.

Direct all correspondence to: Glenda Andrews, School of Humanities and Social Sciences, Charles Sturt University, PO Box 588, Wagga Wagga, NSW, Australia 2678
<gandrews@csu.edu.au>.

5 years (Blevins-Knabe, 1987a, 1987b; Chapman & Lindenberg, 1988, 1989, 1992; Halford & Kelly, 1984; Kallio, 1982; Youniss & Murray, 1970). For reviews see Breslow (1981), Halford (1989), and Thayer and Collyer (1978). In this paper we present evidence relevant to two potential sources of difficulty in transitive inference. One is the structural complexity of the information that must be processed, defined as the number of entities related in the same decision. The second is the need to process content-independent representations.

The Difficulty of Transitivity

Several explanations for the difficulty of transitivity have been proposed. Piaget (1957) claimed that children could not make transitive inferences until they reached the concrete operational stage at 7 or 8 years of age. Chapman (1987, 1990) proposed that transitivity involved assigning values to three relational variables simultaneously. Each value assignment was assumed to require one unit of processing capacity, making the total capacity demand of transitive reasoning equal to three units, which were not available until 7 or 8 years of age.

There is now reasonable consensus that transitivity entails integrating premise elements into an ordered set (Halford, Smith, Dickson, Maybery, Kelly, Bain, & Stewart, 1995; Sternberg, 1980a, 1980b; Trabasso, 1975, 1977; Trabasso, Riley, & Wilson, 1975). Thus premises such as, "John taller-than Peter" and "Peter taller-than Mary" would be integrated into an ordered triple "John taller-than Peter taller-than Mary". Both Gentner, Rattermann, Markman and Kotovsky (1995) and Halford et al. (1995) have recognized that knowledge of monotonically ordered sets is important in this performance.

However, integration of premises into a monotonically ordered triple (a ternary relation) entails a degree of complexity that might be relevant to the difficulty of transitivity. If we consider only one relation, say, "John taller-than Peter," the ordinal position of each entity is ambiguous. John's position could be either tallest or middle. It is only when we also consider "Peter taller-than Mary", that John can be unambiguously assigned to the tallest position. Similarly for the other entities. Ordering the three premise elements in this way requires two binary relations to be considered in a single decision, and this imposes a processing load on children (Halford, Maybery, & Bain, 1986; Rogers & Berch, 1996) and adults (Maybery, Bain, & Halford, 1986).

An alternative view explains the errors of children under 5 years in terms of methodological inadequacies that have resulted in false negative assessments. Such inadequacies have also been responsible for false positive assessments (Breslow, 1981; Thayer & Collyer, 1978). One source of false positives involves use of a labeling strategy that is possible when a 3-element series (A, B, C) with two premise pairs ($A > B$, $B > C$) is used. A is labeled "greater than", and C is labeled "less than" in the premises. Merely repeating these labels yields the correct answer to the inferential question $A ? C$. In a 5-element series, there are four premise pairs (AB, BC, CD, DE), and the inferential question is $B ? D$. Labeling

effects are neutralized because B and D are each labelled "greater than" and "less than" in the premises.

False negative errors can occur if children do not retain the premise information long enough to make the inference. This was investigated by Bryant and Trabasso (1971). They used a 5-element task and children were trained to criterion on the four premise pairs ($A > B$, $B > C$, $C > D$, $D > E$) and ($E < D$, $D < C$, $C < B$, $B < A$), and then tested on all pairwise relations. The critical comparison referred to B and D. Results suggested that 3- to 4-year-olds could make transitive inferences provided they could remember the premise information. However, the training in which pairs were presented initially from shortest to longest, or vice versa, might have unduly facilitated construction of the correct ordering. This assistance, and the elimination of children who failed to learn the premises, would have positively biased the results (Halford, 1992). Subsequent studies avoided these biases (Halford & Kelly, 1984; Kallio, 1982) and found no evidence of transitive inference in children under 5 years.

An alternative way to ensure memory for premise information is to provide a memory aid. One such study (Halford, 1984) used a memory board containing pairs of colored pegs to code the relative lengths of sticks placed inside colored tubes. The colors of the pegs matched the tubes. The relation "left of" in the memory board corresponded to "longer than" for the sticks inside the tubes. For example, if the red peg was left of the green peg in the memory board, then the stick inside the red tube was longer than the stick inside the green tube. The task entailed ordering the tubes from left to right according to the lengths of the sticks inside them. Halford found that 3- to 4-year-olds succeeded when correct ordering depended on processing one relation at a time. They could order two objects, B and C, on the basis of a single relation, B left of C. They could also add a third object, D, to an already-ordered pair provided this could be done by considering only one relation, C left of D. They did not succeed when correct ordering depended on processing two relations in the same decision (e.g., given B left of C, and C left of D, they failed to order objects B and D). A possible reason is the processing load imposed by integration of relations.

Pears and Bryant (1990) provided evidence that seems to contradict Halford's (1984) finding. In their procedure premise relations were represented by pairs of colored blocks arranged into 2-block towers. Participants who were aged from 4 years, 0 months to 4 years, 11 months (mean age 4;6) were required to build 3-, 4-, 5-, or 6-block towers which preserved the spatial relations of the colors in the premises. Before constructing each tower, they predicted which of two colors would be in the higher position in the tower. In the tower-of-5 condition there were four premise towers: A above B; B above C; C above D; D above E. The construction task required children to build a 5-block tower with a top-down order of A, B, C, D, E. Before constructing the tower, children predicted whether B or D would be higher up in the tower. The prediction question required coordination of two premise relations, B above C, and C above D, to produce the tran-

sitive inference B above D. Contrary to Halford's (1984) findings, 4-year-olds succeeded in making transitive inferences about spatial position.

Pears and Bryant (1990) acknowledged that children may have made the inferences by manipulating images of the premise towers, that is by imagining the 'B above C' pair sitting on top of the 'C above D' pair, making it apparent that B was above D. This type of processing was less likely in Halford's (1984) study, because the premise relations were presented in a coded form. "Left of" in the premises represented "longer than" in the series. Children had to map the spatial relations between the pegs in the memory board onto the length relation required by the ordering task, and this may have been the cause of the extra difficulty.

Mapping Between Domains

It seems appropriate to test whether young children can make transitive inferences with different contents, such as placing colored sticks in left-right order. Furthermore a flexible reasoning process should be able to operate across contents, mapping from sticks to blocks, and from the "left-of" relation to the "above" relation, and vice versa. Simple imagery alone would not be sufficient for inferences that involve two different relations. Here rather complex imagery, or a representation that is independent of the particular dimension, is required.

Mapping from sticks to blocks entails a form of analogical reasoning. There is a growing recognition of the central importance of analogy in the reasoning of adults (Clement & Gentner, 1991; Holyoak & Thagard, 1995) and of children (Goswami, 1991; Halford, 1992). Analogy involves a mapping from one representation, the source, to another representation, the target (Gentner, 1983). It is a process that has the generality that imagery lacks, because mappings can be made between different materials or different domains. Young children can perform analogies that depend on mapping single, binary relations in familiar domains, but their ability to integrate binary relations, or process ternary relations, is in dispute (Goswami, 1991; Halford, 1993; Halford, Wilson, & Phillips, in press).

The primary aim of Experiment 1 was to assess young children's ability to make transitive inferences, using the child-appropriate procedure of Pears and Bryant (1990). The processing load imposed by integrating two binary relations into a ternary relation was expected to make the task difficult for young children. The second aim was to assess the robustness of Pears and Bryant's findings with different materials. The third aim was to test the flexibility of transitive inference in young children by requiring mapping from premises in one domain (e.g., sticks) to another (e.g., blocks) or vice versa.

EXPERIMENT 1

The procedure was an extended version of Pears and Bryant's (1990) Tower-of-5 condition, because a 5-term series is the smallest that precludes labeling strategies. There were two nonmapping tasks, nonmapping blocks (B1-B1) where the

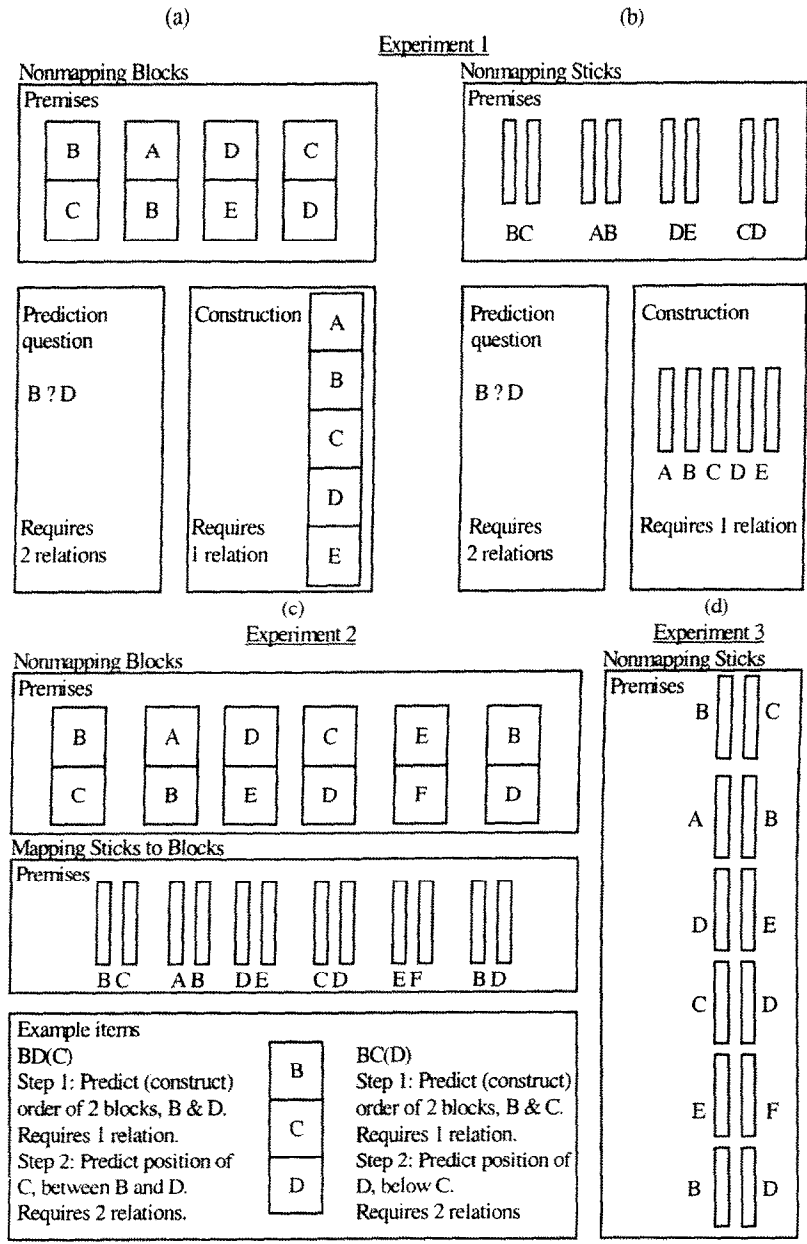


Figure 1. (a) Nonmapping blocks; (b) nonmapping sticks tasks used in Experiment 1; (c) nonmapping and mapping tasks used in Experiment 2; (d) columnar arrangement of sticks used in Experiment 3.

premises were blocks and the children ordered blocks, and nonmapping sticks (St-St), where premises were sticks and children ordered sticks. These are shown in Figures 1a and 1b respectively. The BI-BI task was comparable to Pears and Bryant's Tower-of-5 condition. Children used premise information to predict the relative vertical position of blocks B and D (prediction), and then to construct a 5-block tower with a top-down order A, B, C, D, E (construction). The St-St Task, shown in Figure 1b, was isomorphic to the BI-BI Task. Sticks were substituted for blocks, and the left-right relation was substituted for the above-below relation. Children used the premise information to predict the left-right position of sticks B and D (prediction), and to construct a left-right 5-stick array (construction).¹

There were two mapping tasks, mapping sticks-to-blocks (St-BI) and mapping blocks to sticks (BI-St) in which the premise materials and relations differed from the materials and relations used in the prediction and construction subtasks. In the St-BI task, premise information consisted of pairs of sticks as in the St-St task. Children predicted the relative vertical positions of two blocks, B and D (prediction), and constructed a 5-block tower with top down order A,B,C,D,E (construction). The left-right relation in the premises had to be mapped onto the above-below relation in the prediction and construction subtasks. The BI-St task was similar except that mapping was in the opposite direction.

Based on the earlier findings (Halford, 1984; Pears & Bryant, 1990), we expected 4-year-olds to succeed on prediction questions of the BI-BI and St-St tasks, but their reasoning might not be sufficiently flexible to map from blocks to sticks or vice versa (BI-St, St-BI tasks). The 6-year-olds were expected to succeed on BI-BI, St-St, BI-St, and St-BI tasks.

Method

Participants. Twelve children were recruited from a denominational primary school and twelve were recruited from a preschool in a lower middle class suburb of Gold Coast, Australia. The older group consisted of 5 males and 7 females with a mean age of 5 years, 11 months (range 5;5 to 6;6). The younger group consisted of 6 males and 6 females with a mean age of 4;2 (range 3;9 to 4;7). Race was not recorded but participants were generally of Caucasian descent. Parents' written consent for their children's participation was obtained.

Materials. There were 15 wooden blocks, three each of five colors, red, green, blue, yellow, and white. Each block measured 5 cm × 5 cm × 5 cm. There were 15 wooden sticks, three each of the above colors. Each stick measured 15 cm in length and approximately .5 cm in diameter.

Design. A mixed design was employed. Age (two levels) was the only between-groups factor. The four tasks: nonmapping blocks (BI-BI), nonmapping

¹The tasks described here as nonmapping would be described by Gentner and Toupin (1986) as literal similarity tasks.

sticks (St-St), mapping blocks to sticks (Bl-St), and mapping sticks to blocks (St-Bl) were performed by all participants. Children within each age group were randomly assigned to one of four task presentation orders with the restriction that there were equal numbers in each sequence. The sequences were: Bl-Bl, St-St, Bl-St, St-Bl; St-St, Bl-Bl, Bl-St, St-Bl; Bl-Bl, St-St, St-Bl, Bl-St; and St-St, Bl-Bl, St-Bl, Bl-St. Nonmapping tasks were administered prior to the mapping tasks because they formed the basis of the mapping tasks, and it was necessary to ensure participants' familiarity with them before proceeding to the mapping tasks.

The construction and prediction subtasks were embedded within each of the four tasks. In Pears and Bryant's (1990) experiment, and in the Bl-Bl and St-St tasks of the present experiment, the same set of premise pairs was used for the construction and prediction. This minimized the number of trials required. However because the prediction question involved predicting the order of elements used in construction, prediction necessarily preceded construction. For the Bl-St and St-Bl tasks, it was necessary to ensure that children grasped the correspondence between the relation in the premises and that required by the task. This was best achieved by presenting construction prior to prediction. A greater number of trials was necessary because different sets of premise pairs were now required for construction and prediction.

Procedure

Color names were used throughout to refer to the blocks and sticks. For ease of explanation, letter names A, B, C, D, E will be used here to refer to the blocks according to their top-down order, and to the sticks according to their left-right order. All five colors were used in each series and colors were randomly assigned to ordinal positions for each trial of each task. The premise pairs were always arranged in random spatial order on the table at which the participant was seated. The order of mention of the two blocks or sticks to be ordered was also randomized. Whenever a prediction question was asked, the relevant blocks or sticks were placed in view of the participants so that nonverbal replies could be offered if the children preferred.

Four premise pairs of blocks or sticks representing four adjacent relations (AB; BC; CD; DE) in the 5-element series (A, B, C, D, E), were presented on each trial.

Bl-Bl Task. Children were told that they would be playing a game with blocks. The game involved building a tower using five blocks (construction), and answering questions about some of the blocks (prediction).

Four practice trials were administered. On the first practice trial, the four premise pairs of blocks were presented simultaneously as in Figure 1a. Children were given five blocks, one of each color, and asked to construct a 5-block tower consistent with the order of the blocks in the premises. They were told, for example, that if red was above green in the small towers, red must also be above green in the big tower. Guidance and feedback was given as required. Once the tower

had been correctly constructed, it was removed and the participant's attention was directed to the premises. Three questions were asked about the relative positions of two adjacent pairs of blocks (B?C, C?D) and one nonadjacent pair (B?D). The question was of the form, "When you build the tower, which block will be higher up, B or C?" After answering the questions, children constructed the tower. On the three subsequent practice trials the initial construction of the tower was omitted. The three prediction questions were asked before children constructed the tower. Six test trials were then administered. Premise information was presented simultaneously and was continuously present while children responded to one prediction question, B?D, before constructing the 5-block tower.

St-St Task. Children were told that they would be playing a game with sticks. The game involved placing five sticks in left-to-right order (construction), and answering questions about some of the sticks (prediction). A stuffed toy frog was placed on the left side of the table and the phrase "closer to Froggie" was used instead of "left of" in all instructions and questions. As in the BI-BI task, there were four practice trials followed by six test trials.

The mapping tasks (BI-St, St-BI) each included 2-element construction, 5-element construction, and a B?D prediction question. There were six trials for each subtask. Two-element construction provided practice at mapping. Throughout the mapping tasks, children were reminded of the correspondence between the relation in the premises and the relation required by the task.

BI-St Task. Except for the 2-element construction task, premise information consisted of four pairs of blocks as in the BI-BI task (Figure 1a). Children used this information to order (construction) or predict the order of the sticks (prediction).

In 2-element construction, single pairs were mapped. A pair of blocks was presented say, green above yellow. Blocks-sticks mapping was explained as follows, "If the green block is above the yellow block, then the green stick must go closer to Froggie than the yellow stick." They then arranged the two sticks. Feedback was given after each pair. In 5-element construction, participants arranged five sticks in a left-right order consistent with the premises. In prediction questions, children used the premises to predict the left-right position of two sticks, B and D. The question was of the form, "When you place the sticks, which one will be closer to Froggie, B or D?" This required participants to make transitive inferences.

St-BI Task. This was similar to the BI-St task except that mapping was in the opposite direction. The premises consisted of pairs of sticks (Figure 1b) and the children used this information to make predictions and constructions using blocks.

Results

When assessing transitive inference it is important to ensure that the task could not have been performed by guessing or by processing a single binary relation.

Table 1. Means (and Standard Deviations) for Each Age Group in the Nonmapping Condition of Experiment 1

Age Group	Blocks		Sticks	
	Construction	Prediction	Construction	Prediction
Younger	5.58 (0.67)	3.75 (1.22)	4.83 (1.40)	2.83 (1.70)
Older	5.92 (0.29)	4.25 (0.75)	5.67 (0.65)	4.42 (1.24)

Note. Maximum score = 6.

For example, given $B > C$, $C > D$, correct responses could be achieved 50% of the time by processing either premise alone. The calculation of chance and one-relation baselines is given in the Appendix.

In Experiment 1, prediction preceded construction in the nonmapping tasks, and the order was reversed in the mapping tasks, therefore data from nonmapping and mapping tasks were analyzed separately.

Nonmapping Tasks. Correct responses (out of 6) were subjected to a $2(\text{age}) \times 2(\text{construction-prediction}) \times 2(\text{blocks-sticks})$ mixed analysis of variance (ANOVA). Means and standard deviations are shown in Table 1.

There was a significant effect of age, $F(1, 22) = 11.59, p < .01$. The older group ($M = 5.06$) outperformed the younger group ($M = 4.25$). There was a significant effect of blocks-sticks, $F(1, 22) = 4.41, p < .05$. The sticks tasks ($M = 4.44$) were more difficult than the blocks tasks ($M = 4.88$). The construction-prediction effect was significant, $F(1, 22) = 85.2, p < .001$. Prediction ($M = 3.81$) was more difficult than construction ($M = 5.50$), but this effect was confounded with order of presentation (prediction preceded construction). There were no interactions.

Both age groups performed well on the construction subtasks. Group means for the B?D prediction questions were compared to chance- and one-relation baselines each of which correspond to a probability correct of .50. Two-tailed significance tests are reported except where indicated. The older group's mean scores for the two prediction questions were above the baseline (3.0), $t(11) = 5.73, p < .01$ for blocks prediction, and $t(11) = 3.96, p < .01$ for sticks prediction. The younger group's mean for blocks prediction was marginally above the baseline, $t(11) = 2.13, p < .05$ (one-tailed). The younger group's performance on sticks prediction was not significantly different from the baseline, $t(11) = -0.35, p > .05$.

Mapping Tasks. Correct responses (out of 6) were subjected to a $2(\text{age}) \times 2(\text{direction of mapping}) \times 2(\text{construction-prediction})$ mixed ANOVA. Means and standard deviations are shown in Table 2. Because there were no significant main effects or interactions involving direction of mapping, scores are collapsed across this variable.

Table 2. Means (and Standard Deviations) for Each Age Group in the Mapping Condition of Experiment 1

Age Group	Construction	Prediction
Younger	10.25 (1.42)	5.42 (1.93)
Older	11.83 (0.39)	8.67 (1.30)

Note. Maximum score = 12 after collapsing across direction of mapping.

There were significant main effects of age, $F(1, 22) = 37.76, p < .001$, and of construction-prediction, $F(1, 22) = 99.00, p < .001$. These main effects were modified by a significant Age $\times$ Construction-prediction interaction, $F(1, 22) = 4.30, p < .05$. The construction-prediction effect was significant for the older group, $F(1, 11) = 67.31, p < .001$, and for the younger group, $F(1, 11) = 46.96, p < .001$. The age effect was significant for both construction, $F(1, 22) = 13.84, p < .01$, and prediction, $F(1, 22) = 23.40, p < .001$. The interaction reflects the fact that the age difference is larger for prediction (3.25) than for construction (1.58). On prediction, the older group performed significantly above the baseline, $t(11) = 7.10, p < .01$, however the younger group did not, $t(11) = -1.04, p > .05$.

Discussion

There is equivocal support for Pears and Bryant's (1990) findings in that 4-year-olds' performance was marginally above baseline level on prediction in the BI-BI tasks, being significant only by one-tailed test. However on St-St prediction, performance was at chance level, suggesting that Pears and Bryant's (1990) finding might not be robust to variations in materials or relations. This issue is assessed further in Experiment 3. In the mapping tasks the 4-year-olds performed at baseline level on prediction, suggesting that they cannot perform transitive inferences when they have to map premise information from one display to another. Their performance may reflect reliance on content-specific, imaginal representations.

In the nonmapping tasks in Experiment 1, prediction always preceded construction, as in Pears and Bryant's (1990) study, because this helped keep the procedure simple. In the mapping tasks, construction always preceded the prediction questions, and mapping always followed nonmapping. Order of presentation can be controlled, albeit at the cost of more procedural complexity, and this will be done in Experiment 2.

For the mapping tasks, both age groups performed well on construction, suggesting that the difficulty is not just in mapping from one spatial dimension to another, but is due to the processing required by prediction. Construction could be performed correctly by mapping single relations sequentially. For example, in St-St construction, to place one pair of blocks, say C above D, one premise relation, C left of D, must be mapped. Adding a third block, B, to the pair already placed,

requires one relation, B left of C to be mapped. The resulting top-down order is B, C, D. The 5-block tower can be constructed by mapping the relations sequentially. In contrast, B?D prediction required two relations, B left of C and C left of D, to be mapped in the same decision. The 4-year-olds experienced difficulty in the latter case. Number of relations that need to be processed in a single decision is found to be a major determinant of complexity in a wide range of contexts (Halford et al., in press). In Experiment 2 this factor will be manipulated independently of construction-prediction.

EXPERIMENT 2

The purpose of Experiment 2 was to investigate children's performance on one- and two-relation tasks under four different conditions formed by the combinations of mapping-nonmapping and construction-prediction.

Design of the Task

Three types of items that require two relations to be considered in a single decision were used.

Ordering Nonadjacent Elements. (e.g., order BD, using premises $B > C$, $C > D$). This is the traditional test for transitivity, as used in Experiment 1 (see Figure 1a).

Insertion Between Nonadjacent Elements. (e.g., insert C between B and D, using premises $B > C$, $C > D$, see Figure 1c). This has been commonly used in the transitivity and seriation literature (e.g., Piaget, 1957).

Adding an element to the end of a series, using both adjacent and nonadjacent relations. For example, suppose premises $B > C$, $C > D$ and $B > D$ have been presented, and BC have been ordered. The problem now is to place D to produce the triple BCD. Two premises indicate possible positions for D: $B > D$ would yield the orders BDC or BCD, while $C > D$ would yield BCD. The conflict concerning the correct placement of D can be resolved by considering $B > D$ and $C > D$, that unequivocally yields BCD. $C > D$ may seem more appropriate to us, but this is because we already know the correct ordering. More detail about this type of item is given in the Appendix (see item DE(C)).

This type of item utilizes premises representing relations between nonadjacent elements, such as $B > D$. Such items have not been commonly used in the literature, but were used by Halford (1984), and they increase ecological validity of the task. In real world ordering tasks, one is often confronted with information regarding nonadjacent relations. Items to be rank ordered are not presorted into adjacent pairs. If rank ordering students, the fact that Jane is ahead of Jim does not mean they are adjacent. It is possible that Denise is behind Jane but ahead of Jim. This makes the ordering task more difficult, because the insertion of an interne-

Table 3. Number of Relations Required on Each Step for the Items in Experiments 2 and 3

NA Relation Supplied	Item types				
	Ordering NA Elements ^a	Insertion Between NA Elements	Adding Element to End of Series		
CE	BD(C)	CE(D)	DE(C)	CD(B)	BC(D)
Step 1	2	1	1	—	—
Step 2	—	2	2	1	1
BD	CE(D)	BD(C)	BC(D)	CD(E)	DE(C)
Step 1	2	1	1	—	—
Step 2	—	2	2	1	1

Notes. Each item consisted of two steps. Elements added on Step 2 are shown in parentheses. Dashes indicate steps that were not used. NA = nonadjacent.

^a Two instances of the cases in this column were included.

diate element requires integrating two relations (Jane ahead-of Denise, Denise ahead-of Jim). Attention to just one of these relations does not guarantee correct placement of the intermediate element.

Using these three item types gives a more varied way of assessing children's ability to consider two relations in a single decision, reducing reliance on a single item type. Each of the two-relations item types can be compared with one or more one-relation types of items.

Premise information was similar to Experiment 1 except that a nonadjacent relation, either BD or CE was added, and the series was expanded to six elements, A, B, C, D, E, F. The premise pairs were therefore AB, BC, CD, DE, EF and either BD or CE (see Figure 1c). If end elements, A and F, are excluded to preclude use of a labeling strategy, two subsets, BCD and CDE were available for testing. Table 3 summarizes the items selected for inclusion in the study and the number of relations involved at each step. Letters indicate the blocks ordered with the block added on Step 2 shown in parentheses. Items were selected so that there were equal numbers of one- and two-relation items evenly distributed across steps 1 and 2.

There were four task conditions: nonmapping construction, (Bl-BI:Constr); nonmapping prediction, (Bl-BI:Pred); mapping construction, (St-BI:Constr); and mapping prediction (St-BI:Pred). The nonmapping (mapping) conditions were similar to the Bl-BI (St-BI) tasks in Experiment 1, except for the modifications to premise information and items as noted above. Furthermore, in the current study, construction-prediction was not confounded with number of relations. One- and two-relation items shown in Table 3 were administered in each of these four conditions to two age groups of children.

Predictions

It was predicted that the structural complexity of the task would affect performance. Specifically, steps that required integration of two relations would be more

difficult than those that required a single relation to be considered. This effect was expected to be greater for the younger children, either because of age-related capacity limitations that impose limits on the structural complexity of the concepts they can represent, or because they lack the requisite knowledge for integrating relations. The effect for number of relations was expected to be greater in the mapping condition than in the nonmapping condition.

Method

Participants. There were two groups with 12 participants in each. Participants were from the same denominational school and preschool described in Experiment 1. None of the children who participated in Experiment 1 took part in Experiment 2. The older group consisted of 7 males and 5 females with a mean age of 5;11 (range 5;5 to 6;3). The younger group consisted of 3 males and 9 females mean age of 4;2 (range 3;7 to 4;6.) Participants were of Caucasian descent. Parents' written consent for their children's participation was obtained.

Materials. The materials were similar to Experiment 1 except that an extra color (brown) was added. There were 24 wooden blocks, four of each color. As in Experiment 1, 18 wooden sticks, three of each color were used.

Design. A mixed design was employed. Age (two levels) was a between-group factor. Mapping-nonmapping, construction-prediction, and number of relations (1, 2) were within-subject factors. Six children in each age group received the nonmapping tasks prior to the mapping tasks and six received the reverse order. Half the children received the construction tasks before the prediction tasks, the remainder received the reverse order. The items in Table 3 were presented in each of the four task conditions, nonmapping construction (Bl-Bl:Constr), nonmapping prediction (Bl-Bl:Pred), mapping construction (St-Bl:Constr), and mapping prediction (St-Bl:Pred). The order of presentation of the items within task conditions was randomized.

Procedure

The six premise pairs of blocks (nonmapping condition) or sticks (mapping condition) represented five adjacent relations (AB, BC, CD, DE, EF) and one nonadjacent relation (either BD or CE).

Preliminary testing was conducted to ensure that children understood color names and the relational terms, above and below. Participants were given six blocks, one of each color, and asked to construct a tower, and then to point to two blocks. They were asked which of the two blocks was higher up/lower down in the tower. Adjacent and nonadjacent pairs were tested with order of mention of the blocks randomized.

Training for the nonmapping condition was given immediately prior to the nonmapping tasks. To re-check understanding of the above-below relation, six pairs of blocks were presented. Children were asked to point to a pair of blocks nominated by the experimenter and to state (or otherwise indicate) which block was higher up (or lower down). All six pairs were tested, with feedback given after each pair.

To ensure that children understood that the order of the blocks in the premises must correspond to their order in the completed series, participants constructed a 6-block tower under the experimenter's guidance. Assistance was provided if children experienced difficulty placing any block under those already in place, but only after children had clearly indicated its position. Such assistance was not usually required in the testing phase where only three blocks were used. Children's attention was repeatedly directed to the premises to determine the correct order and to check for inconsistencies in ordering. Once constructed, the tower was placed to one side and the premises were removed.

To help ensure that the children realized that the premise pairs represented relations in the tower, they reconstructed the premises from information contained in the 6-block tower. They were shown two blocks (e.g., B and C) and asked which block should be higher up in the small (premise) tower. The premise tower, B above C, was then constructed. The previously constructed 6-block tower was in full view, and children's attention was directed toward it if necessary. This procedure was repeated until six relations, including the nonadjacent relation (BD or CE), were represented in the two-block premises. Feedback was given after each pair. Inconsistencies were pointed out and corrected. Particular emphasis was given to the pair representing the nonadjacent relation.

In the subsequent construction and prediction tasks in the nonmapping condition, six premise pairs of blocks were presented as shown in Figure 1c.

Bl-Bl:Constr Task. In each of the six items, participants constructed a three-block tower. On Step 1, two blocks were placed. On Step 2, a third block was added to the existing pair. For example, on Step 1 of BC(D), children were given B and C and asked to, "Put these in the right order". They were reminded to, "Look at the small towers to find out which way they go". Errors were corrected before Step 2. On Step 2, they were given D and asked to "Put this one in its correct place" and were again reminded to consult the premises.

Bl-Bl:Pred Task. Children predicted the order of three blocks from the premise information. On Step 1, they predicted which of two blocks would be higher up. On Step 2, they predicted the position of the third block relative to the existing pair. Children were asked not to handle the blocks until they had made their prediction. Item CD(B) will be described. On Step 1, participants were asked, "Which block will be higher up, C or D?" The experimenter showed the two blocks to the subject and then placed them side by side, in random order, on the table in view of the subject. Participants were reminded to consult the pre-

mises to determine the order. Children were credited with success if they stated the color of the correct block or if they pointed to it. After responding, the child placed the blocks in order. Errors were corrected prior to Step 2. On Step 2, the experimenter held up B and asked, "Where should this one go?" B was then placed on the table in view of the subject. They were credited with success on this step if they stated or otherwise indicated a) the correct top-down or bottom-up order of the three blocks; b) that an end block would be higher or lower than the middle block (e.g., B would be higher than C, or D would be lower than C); c) that a middle block would be between (or in the middle of) two blocks; or d) that a middle block would be higher than one block and lower than another. The third block was then placed in position.

Training for the mapping tasks was conducted to ensure that children could identify the stick in each premise pair that was left of the other stick (i.e., closer to Froggie), to ensure that children could use the premise information (sticks) to construct two-block towers, and to predict the relative position of blocks in a two-block tower. Training for the St-BI:Constr task was identical to the St-BI single pairs construction task in Experiment 1. Training for the St-BI:Pred task was similar except that participants predicted the relative vertical position of two blocks rather than placing them in position. In the next phase of training, six premise pairs of sticks were presented and children constructed a six-block tower with the experimenter's guidance. The premise pairs were removed. The tower remained in view while the child reconstructed the premises in a manner similar to the nonmapping training.

Test Phase of the Mapping Condition. In the St-BI:Constr and St-BI:Pred tasks, six premise pairs of sticks were presented. In all other respects, the procedure was identical to the nonmapping condition.

Results

Correct responses were subjected to a 2 (age) $\times$ 2 (nonmapping-mapping) $\times$ 2 (construction-prediction) $\times$ 2 (number of relations) ANOVA. There were no significant main effects or interactions involving construction-prediction. There was a significant effect for nonmapping-mapping, $F(1, 22) = 6.02, p < .05$. The nonmapping tasks ($M = 13.42$) were less difficult than the mapping tasks ($M = 12.50$). There were significant effects for age, $F(1, 22) = 16.65, p < .001$; and number of relations, $F(1, 22) = 97.27, p < .001$. These main effects were modified by a significant Age $\times$ Number of Relations interaction, $F(1, 22) = 17.34, p < .001$. Means and standard deviations are shown in Table 4.

Simple effects analysis showed significant effects of number of relations for the 6-year-olds, $F(1, 11) = 14.87, p < .005$, and the 4-year-olds, $F(1, 11) = 108.31, p < .001$. There was a significant effect of age, $F(1, 22) = 23.00, p < .001$, for two-relation tasks, but not for one-relation tasks, $F(1, 22) = 1.11, p > .05$.

Table 4. Means (and Standard Deviations) for Each Age Group for One- and Two-relation Steps in Experiment 2

Age Group	Number of relations	
	One	Two
Younger	14.58 (1.17)	9.25 (2.01)
Older	15.08 (1.17)	12.92 (1.73)

Note. Maximum score = 16 after collapsing across construction-prediction and mapping-nonmapping.

Chance and one-relation baselines for Steps 1 and 2 combined were .415 (3.32 out of 8) and .563 (4.50 out of 8) respectively (see Appendix for details). Differences between mapping and nonmapping conditions were predicted, therefore performance in comparison to the baselines was evaluated separately for each condition.

Nonmapping Condition. For the one-relation items (construction and prediction combined), the means for the older ($M = 7.75$, $SD = 0.45$) and younger groups ($M = 7.42$, $SD = 0.67$) were above chance, $t(11) = 34.08$, $p < .001$; and $t(11) = 21.13$, $p < .001$, respectively. For the two-relation items, the means for the older ($M = 6.50$; $SD = 1.17$) and younger groups ($M = 5.17$; $SD = 1.12$) were above the chance baseline, $t(11) = 9.41$, $p < .001$, and $t(11) = 5.78$, $p < .001$, respectively. The older group's mean was above the one-relation baseline, $t(11) = 5.92$, $p < .001$. The younger group's mean was marginally above the one-relation baseline, $t(11) = 2.09$, $p < .05$ (one-tailed).

Mapping Condition. For the one-relation items, the means for the older ($M = 7.33$; $SD = 0.99$) and younger groups ($M = 7.17$; $SD = 0.72$) were above chance, $t(11) = 19.33$, $p < .001$, and $t(11) = 14.12$, $p < .001$, respectively. For the two-relation items, the older group's mean of 6.42 (1.16) was above chance, $t(11) = 9.25$, $p < .001$, and one-relation baselines, $t(11) = 5.73$, $p < .001$. The younger group's mean of 4.08 (1.56) was not significantly different from either baseline.

Discussion

Results supported the prediction that structural complexity would influence performance. Across all four task conditions, performance was poorer when two relations had to be integrated in the same decision than when a single relation needed to be considered. The older children gave more correct responses than the younger children in all conditions. However, the age difference was greater for the two-relation cases.

In the nonmapping condition, the performance of the younger group on the two-relation cases was above chance level. However, this does not necessarily indicate that they performed transitive inference, because their performance was only mar-

ginally above the one-relation baseline. The corresponding results in the mapping condition suggest that what success is observed in the nonmapping condition may reflect processes such as imagery that are based on content-specific representations. The group mean for the younger children on two-relation mapping was no better than chance. Examination of the data revealed that the poor performance of the younger children on the two-relation mapping items was not the result of extremely poor performance by a few children, but resulted from chance level performance by most of the children. Only three children made more than two correct responses out of a possible four in the two-relation items in the St-BI:Constr and St-BI:Pred tasks. These results are consistent with those obtained in the mapping condition in Experiment 1 and by Halford (1984). The results are not consistent with Goswami's (1992, 1995) view that children under 5 years can reason about two relations in parallel. Note however that, consistent with Halford (1984), both age groups performed well above chance in the one-relation mapping items, showing that their difficulty was not caused by mapping per se.

Performance on tasks using pairs of sticks as premises might have been adversely affected by the display. The random arrangement of stick pairs might mean that relations between pairs sometimes conflict with relations within pairs. For example, if CD were left of BC, D would be left of B and this would conflict with the correct order, BCD. In Experiment 1, performance was worse on St-St than on the BI-BI task, possibly because of confusion arising from the arrangements of the sticks. In Experiment 3, the performance of 4-year-olds on mapping and nonmapping tasks was examined using a display that eliminates this possible artefact.

EXPERIMENT 3

Experiment 2 was repeated but a columnar arrangement of the premise pairs of sticks was used as shown in Figure 1d. Blocks and sticks were used equally often as premise and task materials as in Experiment 1. Only younger children were tested.

Method

Participants. Twelve children enrolled in a child care centre in the same residential area as the preschool and school described in the previous experiments participated. There were 5 males and 7 females with a mean age of 4;2 (range 3;10 to 4;7). Children were of Caucasian descent and none had participated in Experiments 1 or 2.

Materials. The materials were the same as those used in Experiment 2 except that there were four rather than three sticks of each color. The items used in Experiment 2 (see Table 3) were reused in Experiment 3.

Design. A 2 x 2 x 2 design was employed, in which premise materials (blocks, sticks), target materials (blocks, sticks) and number of relations (1, 2) were manipulated within subjects.

Procedure. In Experiment 2, no difference was found between construction and prediction, therefore construction tasks only were used. There were two non-mapping tasks, (BI-BI, St-St), and two mapping tasks, (BI-St and St-BI). Training procedures were modified to include trials where left-to-right arrays of sticks were constructed. When the premise relations were represented by sticks, the pairs of sticks were displayed in a columnar arrangement (see Figure 1d). In all other respects, the procedure was the same as in Experiment 2.

RESULTS AND DISCUSSION

Preliminary analyses showed no significant effects of materials for either the one-relation or two-relation items in the nonmapping condition, and no effect of direction of mapping for the one-relation or two-relation items. Therefore these variables were not included in subsequent analyses. Means and standard deviations are shown in Table 5.

Correct responses were subjected to a 2 (mapping-nonmapping) $\times$ (2 (number of relations) repeated measures ANOVA. There were significant main effects of nonmapping-mapping, $F(1, 11) = 12.57, p < .01$, and number of relations, $F(1, 11) = 126.55, p < .001$, that were modified by a significant Nonmapping/Mapping $\times$ Number of relations interaction, $F(1, 11) = 6.49, p < .05$.

Examination of the interaction showed that the effect of nonmapping-mapping was significant for the two-relation steps, $F(1, 11) = 14.47, p < .01$, but not for the one-relation steps. The effect of number of relations was significant in the non-mapping condition, $F(1, 11) = 91.67, p < .001$, and the mapping condition, $F(1, 11) = 71.95, p < .001$.

For the two-relation steps in the nonmapping condition, the mean was significantly above chance (3.32), $t(11) = 4.47, p < .001$, but not significantly above the

Table 5. Means (and Standard Deviations) for One- and Two-relation Steps in the Nonmapping and Mapping Conditions in Experiment 3

Condition	Number of relations	
	One	Two
Nonmapping	7.42 (0.67)	4.92 (1.24)
Mapping	7.33 (0.89)	3.67 (0.89)

Note. Maximum score = 8.

one-relation baseline (4.50), $t(11) = 1.17, p > .05$. For the two-relation steps in the mapping condition, the mean was not significantly different from chance, $t(11) = 1.36, p > .05$. It was significantly below the one-relation baseline (4.50), $t(11) = 3.23, p < .05$, suggesting that participants resorted to guessing when they were unable to integrate two relations.

The results show that the two-relation steps were more difficult than one-relation steps for 4-year-olds, but this effect was greater in the mapping condition. Performance on the two-relations task was better than chance in the nonmapping condition only. It was not better than the one-relation baseline in either condition, suggesting that the children's performance was not based on premise integration.

Ordering Nonadjacent Elements—Supplementary Analysis

A supplementary analysis was performed of items that required nonadjacent elements to be ordered by considering two relations in a single decision (e.g., ordering B and D by considering $B > C$ and $C > D$). These were contrasted with items in which the same two premises were considered in separate decisions (e.g., where B and C were ordered considering $B > C$, as on Step 2 of CD(B) and where C and D were ordered considering $C > D$, as on Step 2 of BC(D)).

Items involving consecutive processing were scored correct if both steps were performed correctly. Items requiring two premises to be considered in a single decision were scored as correct if the first (and only) step was correct. An ANOVA with nonmapping-mapping and number of decisions as within-subject factors showed a significant effect of number of decisions, $F(1, 11) = 15.68, p < .01$. There was no effect of nonmapping-mapping and no interaction. Consecutive performance ($M = 6.75, SD = 1.22$) was better than when the two relations had to be considered in a single decision ($M = 4.83, SD = 1.40$). This difference obtained despite the fact that the probability of error was greater for the consecutive decision items.

Combined Samples from Experiments 2 and 3

Experiments 1 to 3 provide equivocal data concerning performance of 4-year-olds on the two-relation nonmapping task. Performance was marginally above the one-relation baseline in Experiments 1 and 2, but not in Experiment 3. The two-relation BI-BI:Constr and BI-BI:Pred items in Experiment 2 are equivalent to the two-relation BI-BI items in Experiment 3 (given that there was no difference between construction and prediction in Experiment 2), so it is appropriate to combine the two samples. For the younger group in Experiment 2, scores for the two-relation BI-BI:Pred and BI-BI:Constr tasks (each out of 4) were averaged, and pooled with scores (out of 4) for the two-relation BI-BI task in Experiment 3. The mean (2.46) for the combined samples ($N = 24$) was not significantly greater than the one-relation baseline (2.25), $t(23) = 1.39, p > .05$. The same test combining Experiment 3 with two-relation BI-BI:Constr items only from Experiment 2 yielded $t(23) = 1.15, p > .05$. Therefore, the 4-year-olds have not performed better

than the one-relation baseline in the nonmapping condition. It was inappropriate to combine data from the mapping tasks because of the different sticks displays.

EXPERIMENT 4

Given that relational complexity has been the most potent factor in Experiments 1-3, Experiment 4 was designed to provide a more stringent test of this factor, and to provide a further assessment of 4-year-olds' ability to make transitive inferences. The two-relations items all entailed ordering nonadjacent elements, followed by inserting an element between them (i.e., item BD(C), without presentation of $B > D$, in Table 3).

The procedure was modified in several ways. Five-element series with adjacent relations only were used. The mapping condition was not included. To facilitate administration and to reduce testing time, two-dimensional squares of colored paper were used instead of wooden blocks. Identical premise displays were used for one- and two-relation conditions. Guessing and nontransitive strategies on the two-relation trials were controlled by requiring placement of square C after squares B and D. If the placement of B above D on Step 1 had been performed by integrating premises BC and CD yielding BCD, as well-validated process models indicate (e.g., Trabasso, 1975), then children should already know the correct position of C. Errors on Step 2 would suggest that the earlier response was based on a nonintegration strategy such as guessing or a one-relation strategy.

To avoid confounding number of steps with complexity, the one-relation items also required two steps. Squares B and C (or C & D) were ordered on Step 1 and the remaining square, D (or B) was added on Step 2 (i.e., items BC(D) and CD(B), without presenting $B > D$, in Table 3). Thus, on both the one-relation and two-relation items, children ordered three squares B, C, and D. Relational complexity was manipulated solely by the order in which the squares were placed. Correct ordering of BD followed by correct placement of C on the two-relation items would constitute strong evidence that children were integrating the premise relations, and should help to resolve the question whether young children can perform this task.

Method

Participants. There were 30 children in each of three age groups. Children were recruited from a state-run primary school and two privately run kindergartens in working class and lower middle class suburbs of Brisbane, Australia. The youngest group consisted of 13 females and 17 males with a mean age of 4;3 (range 3;7 to 5;0). The middle group consisted of 18 females and 12 males with a mean age of 5;6 (range 5;1 to 6;0). The oldest group consisted of 13 females and 17 males with a mean age of 6;6 (range 6;1 to 7;0). Participants were primarily of Caucasian descent. Parents' written consent for their children's participation was obtained. Children were screened for color blindness using Plates 26-38 of Ishi-

hara's Test for Colour-Blindness (1980). Only those without color vision deficiencies were included. (Two boys were excluded.)

Design. Age was a between-groups factor with three levels (4, 5, 6). Number of relations was a within-subjects factor with two levels (1, 2).

Materials

Four pairs of colored squares were used in each premise display. Each of the 25 premise displays was mounted on an A4-sized sheet of white paper. Colors were the same as in Experiment 1 except that purple was substituted for white (to provide contrast with the white paper). The squares in each premise pair differed in color and were arranged one above the other with a gap of 0.5 cm between the upper and lower squares. The four premise pairs were placed side by side across the page 5 cm down from the upper edge. Each premise display was used for a separate item: three one-relation practice items; four two-relation practice items; eight one-relation test items; eight two-relation test items; two two-relation filler items. The displays for the one-relation test items and the two-relation test items were identical. Each display was placed inside a transparent cover in a display book. In constructing the displays, controls were incorporated to negate the effect of response biases such as color or position preferences.

Displays for the practice items were constructed so that each color appeared in each ordinal position (A, B, C, D, E) at least once. The four premise pairs each occurred in the four left-right spatial positions at least once. The relative positions of BC and CD were varied. BC was left of CD on four items (2 x one-relation; 2 x two-relation) and right of CD on three items. The degree of separation between BD and CD was varied. BC and CD were adjacent on three trials (1 x one-relation; 2 x two-relation), separated by one premise on two items (1 x one-relation, 1 x two-relation item) and by two premise pairs on two items (1 x one-relation, 1 x two-relation).

Displays for the test and filler items were constructed in a similar manner. Each colour appeared in each ordinal position at least twice. Each of the four premise pairs occurred at least twice in each of the four left-right spatial positions. BC was placed left of CD on four test and one filler trials and right of CD on the remaining trials. The degree of separation between BC and CD for the eight test trials was: zero (three trials); one premise pair (three trials); and two premise pairs (two trials). The separation for the filler trials was zero (one trial) and one premise pair (one trial).

Squares (3 cm x 3 cm) for the task responses were made from the same colored paper as the premises and attached to cardboard squares of the same size.

Procedure

Introduction and Practice Items. Children were tested individually by a female experimenter in a quiet room of their school or kindergarten. The display book was opened but a sheet of paper concealed the premise display. The experimenter showed the participant three coloured squares and explained that the game involved building a tower. She demonstrated by placing the squares one above the other (i.e., nearer the top edge of the display) on the lower half of the page. Having constructed a three-element tower, the experimenter commented, "Some of these squares might be in the wrong place." She then removed the squares and said, "We have to make sure the squares are in the right order. We have some clues to help us". The experimenter then removed the sheet of paper to reveal the premise pairs and explained the relevance of the premises (clues). Pointing in turn to each pair, she explained what it meant. For example, "This clue tells us that yellow must be higher up than blue in our tower. This clue tells" and so on.

The one-relation practice items were presented. The experimenter said, "Now let's see if you can build a tower." The child was given two squares corresponding to positions B and C (or C & D) and told to look at the clues and to work out which one should be higher up in the tower. Once the initial pair was correctly ordered, the experimenter said, "Let's make the tower bigger". The third square was handed to the child who was reminded to look at the clues to determine the correct position of the third square. The experimenter indicated the three options by saying, "It might go up the top, in the middle, or down the bottom" in counterbalanced order of mention. After Step 1 and again after Step 2, the experimenter demonstrated how to check whether the response was correct by referring to the premises, she gave feedback regarding accuracy, and children were encouraged to correct any errors.

The procedure for the two-relation practice items was similar except that children were told that the items were "a bit different" and prompts were added as required. If children commented that the colors were not there (meaning that they were not present in the same premise pair), the experimenter responded, "They are not in the same clue, but you still need to look at the clues. You might have to look at more than one clue." Children who did not respond after a short time were told to guess and the experimenter proceeded to Step 2 (adding C), irrespective of whether the placement of B and D was correct. After the placement of C, the experimenter demonstrated how to check the constructed order against the premises BC and CD. The practice items were administered in the same order to all children.

Test and Filler Items. The test items were of the same form as the corresponding practice items. The structure of four one-relation items was BC(D) and the structure of the remaining four was CD(B). All eight two-relation items took the form BD(C). The test items were administered in a similar manner to the practice items except that no feedback was given and incorrect orderings on Step 1

were not corrected prior to Step 2. In previous experiments, assessment was based on either Step 1 or Step 2 alone (see Table 3) and therefore errors on Step 1 were corrected so as not to contaminate Step 2. However in Experiment 4 assessment was based on both steps jointly. Errors on Step 1 of two-relation items (BD) could not be corrected because placement of C on Step 2 was intended as a check for nontransitive strategies on Step 1. To maintain comparability with the two-relations condition, errors on Step 1 of one-relation items were not corrected. This might inflate the difficulty of one-relation items compared to the previous experiments, and could dilute the complexity manipulation but was necessary to achieve tight experimental control.

The two filler items were of form BD(E) and BD(A). Like the two-relation items they required integration of two premise pairs on Step 1. However the correct position of the third square was either below D or above B, rather than between B and C. It was reasoned that if all two-relation items had the BD(C) form, some children might have realized that whenever the first two elements did not occur in the same premise, the correct position of the third square was between the other two squares. Filler items were included to prevent this response bias developing.

On all test and filler items, the experimenter recorded the positions in which the squares were placed on Steps 1 and 2. Thirty different orders of the 16 test items (8 x one-relation and 8 x two-relation) were generated. These were random with three exceptions: the two items (1 x one-relation, 1 x two-relation) referring to identical displays did not occur in succession; no more than three items of a given complexity were presented consecutively; and filler items were inserted following the third and fifth two-relation trials. Each order was administered to one child in each age group.

RESULTS AND DISCUSSION

Responses to the one-relation items were scored as correct if squares B and C (or C & D) were correctly ordered on Step 1 and D (or B) was correctly placed on Step 2. The two-relation items were scored as correct if B and D were correctly ordered on Step 1 and C was correctly placed on Step 2.

Preliminary analyses conducted to examine for possible effects or interactions involving the nonfocal variables revealed no significant effects nor interactions involving the degree of separation between BC and CD. An ANOVA involving age group, number of relations and side revealed a significant effect of side, $F(1, 87) = 10.55, p < .01$, indicating that items in which BC was right of CD in the display were more difficult than items in which BC was left of CD. A possible interpretation is that children inspected the premises from left-to-right and mentally constructed the tower in a top-down direction, that is A, B, C, D, E. When BC was left of CD, the order in which the premises were encountered matched the top-down order of the elements in the tower. When BC was to the right of CD, a mis-

match would have occurred necessitating the mental rearrangement of the premises before integration. This interpretation is speculative. Its validity is not a major concern for our present purposes because side did not interact with number of relations nor age group. Side was not included in further analyses.

The means and standard deviations for the three age groups are shown in Table 6. A 3 (age group) $\times$ 2 (number of relations) ANOVA revealed a significant effect of age group, $F(2, 87) = 23.70, p < .001$. Scheffe tests showed that the 4-year-olds' mean was significantly lower than the means for the 5- and 6-year-olds. There was a significant effect of number of relations, $F(1, 87) = 92.69, p < .001$. The one-relation items ($M = 5.40$) were easier than the two-relation items ($M = 3.14$). The Age group $\times$ Number of Relations interaction was not significant. As in Experiments 1, 2 and 3 the effect of relational complexity was large, despite the dilution caused by certain aspects of the procedure.

T-tests were used to determine whether the means in Table 6 differed significantly from chance (1.32 correct out of 8), and one-relation (2 correct out of 8) baselines. As in previous experiments, all groups performed above chance on one-relation items. For two-relation items, the 5- and 6-year-olds' means were above chance level, but the 4-year-olds' mean was not, $t(29) = 1.18, p > .05$. The 5- and 6-year-olds' means were also above the one-relation baseline, $t(29) = 5.32, p < .001$ and $t(29) = 6.18, p < .001$ respectively. Binomial tables indicate that individual scores of 4 or more are significantly above chance, and scores of 5 or more are significantly above the one-relation baseline. Table 7 shows the percentage of children in each age group with above baseline scores.

Tables 6 and 7 indicate that performance on the one-relation items was worse than in Experiments 1, 2, and 3. In the current experiment, incorrect placements on Step 1 were not corrected prior to Step 2, as noted under Method. To assess the effect of this an ANOVA with step (1, 2) a within-subjects factor and age group (4,5,6) as a between-groups factor was conducted. It revealed a significant effect

Table 6. Means (and Standard Deviations) for One- and Two-relation Items by Age Group in Experiment 4

Age Group (years)	Number of Relations	
	One	Two ^a
4	3.80 (1.99)	1.73 (1.91)
5	5.67 (2.06)	3.57 (1.61)
6	6.73 (1.89)	4.13 (1.89)

Notes. Maximum score = 8. Chance yields 1.32 correct out of 8 for one-relation and two-relation items.

^a One-relation strategy yields 2 correct out of 8 on two-relation items

Table 7. Percentages of Children with Scores Above Chance for One-relation Items (>3) and with Above Chance (>3) and One-relation Baselines (>4) for Two-relation Items by Age Group

Age Group (years)	Number of Relations		
	One	Two	
	Above Chance	Above Chance	Above One-relation Baseline
4 (<i>n</i> = 30)	43.33	20.00	10.00
5 (<i>n</i> = 30)	80.00	53.33	26.67
6 (<i>n</i> = 30)	90.00	63.33	43.33

of step, $F(1,87) = 30.96$, $p < .001$, indicating that Step 2 ($M = 5.76$) was more difficult than Step 1 ($M = 6.69$). Step did not interact with age group.

Children who were capable of processing a single relation but who erred on Step 1, perhaps because of a momentary lapse in attention, thereby placing C above B (or D above C), might nevertheless have considered a single premise relation on Step 2. To illustrate, suppose C was incorrectly placed above B on Step 1. When given D on Step 2, children would refer to premise CD. Placement of D in middle position constitutes a valid one-relation response when considered independently of Step 1. Step 2 responses were rescored giving credit for placing the third square in middle position following a Step 1 error. The Step 2 mean increased to 6.27. An ANOVA using the new Step 2 scores showed that the effect of step was reduced, though still significant, $F(1, 87) = 5.37$, $p < .05$. The means (standard deviations) for 4-, 5- and 6-year-olds were 5.20 (1.47), 6.30 (1.69), and 7.30 (1.26) respectively.

To assess whether performance reflected difficulties with the procedure, an additional analysis was conducted in which only those children with above-chance scores on the one-relation items were included, because they can presumably cope with the situation. The two-relation means (standard deviations) for the reduced sample ($N = 64$) were 2.39 (2.33); 3.71 (1.57); and 4.15 (1.85) for 4-, 5- and 6-year-olds respectively. An ANOVA showed that the age effect remained significant, $F(2, 61) = 3.98$, $p < .05$. Scheffe test of ranges revealed that the 4-year-olds' mean was significantly lower than the 6-year-olds' mean. *T*-tests showed that the 5- and 6-year-olds' means were above chance level, but the 4-year-olds' mean was not, $t(12) = 1.65$, $p > .05$. The 5- and 6-year-olds' means were also above the one-relation baseline, $t(23) = 5.32$, $p < .001$ and $t(26) = 6.03$, $p < .001$ respectively. Thus after controlling for mastery of the one-relation items, the performance of the 4-year-olds on the two-relation items was not significantly better than either baseline.

Percentages Correct on Steps 1 and 2

To provide a more complete understanding of children's performance, the percentages of children whose Step 1 responses were correct were computed for each

Table 8. Mean Percentage of Children in Each Age Group with Correct Responses to Steps 1 and 2 for One-relation and Two-relation Items

Age Group (years)	One-relation Items		Two-relation Items	
	Step 1	Step 2 ^a	Step 1	Step 2 ^a
4	69.17	68.46	51.67	41.68
5	85.83	83.05	54.17	83.68
6	92.08	91.81	54.58	94.79

Note. ^a Step 2 entries are conditionalized on Step 1. They refer to the percentages of children with correct Step 2 responses given that Step 1 was correct.

item. Of those with correct Step 1 responses, the percentages with correct Step 2 responses were computed. Table 8 shows the percentages after averaging over the eight one-relation and eight two-relation items.

For the one-relation items, the Step 1 and Step 2 percentages within each age group were approximately equal. For the two-relation items, the pattern for the 4-year-olds suggests that correct responses on Step 1 were based either on guessing or on a one-relation strategy each of which yields a probability correct of 0.50. For Step 2, guessing and one-relation strategies yield probabilities correct of 0.33 and 0.50 respectively (see Appendix). If half the children used each strategy, the probability correct for Step 2 given that Step 1 was correct would be 0.42, which is close to the observed level. The high Step 2 performance of the 5- and 6-year-olds suggests that their correct Step 1 responses reflect a two-relation, premise integration strategy, because performance was well above the level expected if nonintegration strategies were employed. Correlations between Steps 1 and 2 were computed separately for each two-relation item and each age group. The mean correlations (ranges) were $r = .29$ (.00 to .65), ns., $r = .61$ (.40 to .73), $p < .01$, and $r = .70$ (.46 to .88), $p < .01$, for the 4-, 5-, and 6-year-olds respectively. Thus Step 1 accuracy was a good predictor of Step 2 accuracy for the 5- and 6-year-olds, but not for the 4- year-olds.

Analysis of Strategies/Errors

The data in Table 7 can be further analyzed by determining the types of responses made over Steps 1 and 2. The two-relation items were examined to determine the types of strategies children used. Responses to each of the eight items were classified according to correctness on Steps 1 and 2. Table 9 shows the patterns across the four cells indicative of four strategies; two-relations, combination of one- and two-relations, one-relation, and guessing. The patterns were derived as follows.

Consistent use of a two-relation strategy would yield perfect performance on BD and C. This would be indicated by a 100% entry in cell D and the zero entries in cells A, B and C. A score of 5 or better is significantly above the one-relation

Table 9. Criteria for Strategy Classification

		Two-relation		Combination		One-relation		Guessing	
		BD		BD		BD		BD	
		0	1	0	1	0	1	0	1
C	0	A —	B —	>2	—	>3	<A	—	>1 ^a
	1	C —	D >4	0	>2	<A	<A	>0 ^a	—

Notes. Criteria used to define strategies were based on frequencies for correct (1) and incorrect (0) placement of BD (columns) and C (rows). Cell A (0,0) indicates incorrect placement to both BD and C. Cell B (1,0) indicates correct placement of BD with incorrect placement of C. Cell C (0,1) indicates incorrect placement of BD with correct placement of C. Cell D (1,1) indicates correct placement of both BD and C. For example, a two-relation strategy was defined if BD and C were placed correctly (cell D) in more than 4 items. Dashes indicate cells not relevant to the criteria.

^a Applied disjunctively

baseline by binomial test, so participants with more than 4 items out of 8 in Cell D were classified as using a two-relations strategy (this is the same criterion as used in Table 7).

Consistent use of a one-relation strategy would yield probability correct of .50 on step 1 (BD). The probability of a correct response for Step 2 (C) differs according to whether BD was correct or incorrect. If BD were correct, C will be correct with probability of .50, yielding 25% correct in cells B and D. If BD were incorrect, the probability correct for C placements is zero. The reason is that if children considered BC only, then C would be placed in bottom position. If they considered CD only, then C would be placed in top position. Thus the expected value for cell A (BD incorrect, C incorrect) is 50%. The expected value for cell C (BD incorrect, C correct) would be 0%. Reliance on a single relation is the only strategy that yields higher frequencies in Cell A than in Cells B, C and D. Participants were classified as using a one-relation strategy if Cell A frequencies were greater than 3, and greater than frequencies for cells B, C and D.

Inspection of detailed records indicated that some participants were using a one-relation strategy on some trials and a two-relation strategy on others. Participants were classified as using this "combination" strategy if frequencies were greater than 2 in both cells A and D, and the frequency of cell C was zero.

Expected frequencies arising from a guessing strategy are based on probabilities of 0.50 for step 1 and 0.33 for Step 2 for correct responses and 0.50 for Step 1 and 0.67 for Step 2 for incorrect responses. Thus the values for cells A, B, C and D are 33.3%, 33.3%, 16.5% and 16.5% respectively. Guessing is the only strategy where predicted frequencies are non-zero in cell C, and where the highest predicted frequency is in cell B. Participants were classified as guessing if *either* their frequency in cell C was non-zero, *or* their frequency in cell B was 3 or more. The classification criteria were applied in the order two-relations, combination, one-relation, guessing (so the criteria for guessing are applied conjunctively with

Table 10. Percentages of Children in Each Age Group Using Each Strategy

Age Group (years)	Strategy type			
	Two-relation	Combination	One-relation	Guessing
4	10.00	10.00	33.33	46.67
5	26.67	26.67	13.33	33.33
6	43.33	13.33	23.33	20.00

absence of criteria for the other three categories). All participants were accounted for.

The percentages of children using each strategy are shown in Table 10. Strategies varied according to age, $\chi^2(6, N = 90) = 15.36, p = .02$. Use of the two-relations strategy increased with age, and 57% of 6-year-olds used either this or a combination strategy. Of the 4-year-olds, 80% used either a guessing or a one-relation strategy. This is consistent with the finding that their group mean is no better than chance, and few of them performed better than the one-relation baseline.

GENERAL DISCUSSION

Number of relations always produced a strong effect. All age groups performed much better on one-relation items, performance never falling below 90% correct in the nonmapping condition in Experiments 1-3. By contrast, performance on two-relation items was much lower, being 55-65% correct for 4-year-olds, and 72-81% in older children. In Experiment 4, performance on one-relation items was lower due to the requirement to be correct on both steps and because step 1 errors were not corrected, but was still much better than the two-relation performances, for all age groups. Number of relations was the largest effect in every experiment where it was analyzed. Furthermore, the data from Experiment 2 supports an interpretation of the construction-prediction manipulation in Experiment 1 as corresponding to processing one versus two relations.

On two-relation tasks older children consistently performed better than chance, but 4-year-olds' performance was at best marginal. Their performance can be attributed mostly to processing one relation, rather than integrating two relations. They were only marginally better than the one-relation baseline for the easiest BI-BI task in Experiments 1 and 2. They were not above the one-relation baseline in the combined nonmapping samples from Experiments 2 and 3, and they were not above either the chance- or one-relation baseline in Experiment 4 where the procedure of requiring C to be inserted after ordering BD, provided more stringent control of nonintegration strategies. This produced lower absolute levels of correct responses, but performances would have been less inflated by guessing than in Experiments 1-3.

The procedure used in Experiment 4 also permitted classification of the strategies used (Table 10). Only 20% of 4-year-olds showed evidence of processing two relations, either alone or in combination with a one-relation strategy, whereas 53% of 5-year-olds and 57% of 6-year-olds did so. The picture that emerges is one of development that is gradual in two respects. First, the number of participants that integrate relations increases gradually over age. Second, participants appear to use four distinct strategies in succession. The modal strategy for 4-year-olds is guessing, with one-relation being the next most frequent. Five-year-olds employ guessing, to a lesser extent, with most participants using either a combination of one- and two-relations strategies, or a two-relations strategy. By age 6 years, the modal strategy is two relations.

Complexity appears to be more important than mapping. The first three experiments indicated that mapping *per se* causes little difficulty when one relation is involved, and there is abundant evidence that young children readily perform simple analogies that involve a single binary relation (Goswami, 1991; Halford, 1993). Thus what seems to develop at approximately 5 years (median age) is the ability to form representations that integrate two binary relations.

How do we reconcile our findings with the claims of Goswami (1995) that 3- and 4-year-olds readily map ternary relations? Children's mappings in Goswami's first experiment might have been based on absolute size rather than on binary or ternary relations (see Halford et al., *in press*). Furthermore, because of the ages of the children in Goswami's second and third experiments, her data do not challenge the proposition that the ability to map ternary relations develops at approximately 5 years of age.

Young children's difficulties do not seem to be attributable to the procedure, that is appropriate to their age, and provides mnemonic aids. This is confirmed by the 4-year-olds' consistently high level of performance on items involving one relation, or where two relations can be processed consecutively. The difference between BI-BI and St-St tasks in Experiment 1 appears to have been due to the nature of the sticks display because the difference disappeared with the improved display in Experiment 3. However the changed display produced only a small improvement, and 4-year-olds' performance on the two-relations items in Experiment 3 was still poor (61.5% correct). The absence of an effect of direction of mapping in Experiment 1, indicates that the sticks display did not adversely affect performance on the mapping tasks, and by inference, cannot explain the results in the mapping condition in Experiment 2. Direction of mapping also had no effect in Experiment 3.

The findings are consistent with those of Halford (1984) and of Markovits, Dumas, and Malfait (1995). Markovits et al. investigated transitivity in 4-, 6-, and 8-year-olds using a different variant of Pears and Bryant's procedure. The procedural differences mean that their results are not directly comparable to ours. However the twelve 4-year-olds who proceeded to the experimental phase in that study, showed no evidence that they could make transitive inferences about spatial

position. Only the 8-year-olds performed reliably better than chance on the majority of items. This is closer to the age that would be predicted from Piaget's theory. Neither the current results nor those of Markovits et al. are consistent with Pears and Bryant (1990), who claimed that 4-year-olds made transitive inferences.

The functional-operational distinction referred to by Chapman and Lindenberg (1988) can be interpreted in terms of the process models used here. They argued that transitivity tasks in which the premise elements are presented simultaneously require only functional reasoning, operational reasoning being required only when premises are presented successively. If all three elements are presented simultaneously in left-right order, the premise integration requirement is obviated, and the associated processing load is not imposed. Therefore simultaneous and successive presentations impose markedly different processing loads. This does not mean that the functional-operational distinction is invalid, but only that it can be subsumed under a processing load account.

The major cause of difficulty with transitive inference appears to be the relational complexity of the task that requires premises expressing binary relations to be integrated into ordered triples (ternary relations). There is an issue as to whether this difficulty is related to knowledge or capacity. Thus Gentner et al. (1995) propose that relational reasoning depends on knowledge of higher order relations. We would not dispute the importance of knowledge, but it does not logically exclude a role for capacity. Halford et al. (in press) have proposed that processing capacity depends on complexity of relations processed in parallel. They argue that this is a general metric that applies to child and adult cognition and to higher animals, and across a wide range of domains. They have shown that integration of premises into ordered 3-tuples, as in our two-relations tasks, entails processing ternary relations, whereas our one-relations tasks entail processing binary relations. Premise integration has been shown to impose a processing load both on adults and children (Halford et al., 1986; Maybery et al., 1986; Rogers & Berch, 1996). A theory that attributes the effects exclusively to knowledge would seem to have difficulty accounting for these effects. The large size of the effects observed in the experiments reported here suggests that relational complexity is the most important factor in children's transitive inferences.

Our data suggest that ability to process ternary relations develops gradually, appearing in only a small minority of 4-year-olds, and with a median age of approximately 5 years, and it is still far from ceiling at 6 years. This finding is supported by Andrews (1996, 1997) who manipulated relational complexity in the domains of transitivity, hierarchical classification, class inclusion, cardinality, sentence comprehension, and hypothesis testing. All tasks loaded on a single factor, relational complexity which accounted for 82% of the age-related variance in fluid intelligence. This suggests that relational complexity has some degree of domain independence. Furthermore the percentage of children that processed ternary relations increased from 15% at age 4, to 41% at age 5, 63% at age 6, 72% at age 7, and 80% at age 8 years. These findings are consistent with the present data

in showing that the ability to process ternary relations develops gradually over the age range 4 to 8 years.

REFERENCES

- Andrews, G. (1996). *Assessment of relational reasoning in children aged 4 to 8 years*. (ERIC Document Reproduction Service No. ED 402 073).
- Andrews, G. (1997). *Relational complexity as a capacity construct in cognitive development*. Unpublished doctoral dissertation, University of Queensland, Australia.
- Blevins-Knabe, B. (1987a). Development of the ability to insert in a series. *Journal of Genetic Psychology*, 148(4), 427-441.
- Blevins-Knabe, B. (1987b). Remembering and constructing an order. *Journal of Genetic Psychology*, 148(4), 453-468.
- Breslow, L. (1981). Reevaluation of the literature on the development of transitive inferences. *Psychological Bulletin*, 89(2), 325-351.
- Bryant, P. E., & Trabasso, T. (1971). Transitive inferences and memory in young children. *Nature*, 232, 456-458.
- Chapman, M. (1987). Piaget, attentional capacity, and the functional implications of formal structure. *Advances in Child Development and Behavior*, 20, 289-334.
- Chapman, M. (1990). Cognitive development and the growth of capacity: Issues in neo-Piagetian theory. In J. T. Enns (Ed.), *The Development of attention: Research and theory* (pp. 263-287). Amsterdam, North Holland: Elsevier Science Publishers.
- Chapman, M., & Lindenberger, U. (1988). Functions, operations, and decalage in the development of transitivity. *Developmental Psychology*, 24(4), 542-551.
- Chapman, M., & Lindenberger, U. (1989). Concrete operations and attentional capacity. *Journal of Experimental Child Psychology*, 47, 236-258.
- Chapman, M., & Lindenberger, U. (1992). Transitivity judgments, memory for premises, and models of children's reasoning. *Developmental Review*, 12, 124-163.
- Clement, C. A., & Gentner, D. (1991). Systematicity as a selection constraint in analogical mapping. *Cognitive Science*, 15, 89-132.
- Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7, 155-170.
- Gentner, D., Ratterman, M. J., Markman, A., & Kotovsky, L. (1995). Two forces in the development of relational similarity. In T. J. Simon & G. S. Halford (Eds.) *Developing cognitive competence: New approaches to process modeling* (pp. 263-313). Hillsdale, N. J: Erlbaum.
- Gentner, D., & Toupin, C. (1986). Systematicity and surface similarity in the development of analogy. *Cognitive Science*, 10, 277-300.
- Goswami, U. (1991). Analogical reasoning: What develops? A review of research and theory. *Child Development*, 62, 1-22.
- Goswami, U. (1992). Commentary. *Human Development*, 35, 222-225.
- Goswami, U. (1995). Transitive relational mappings in three- and four-year-olds: The analogy of Goldilocks and the three bears. *Child Development*, 66, 877-892.
- Halford, G. S. (1984). Can young children integrate premises in transitivity and serial order tasks? *Cognitive Psychology*, 16, 65-93.

- Halford, G. S. (1989). Reflections on 25 years of Piagetian cognitive developmental psychology, 1963-1988. *Human Development*, 32, 325-387.
- Halford, G. S. (1992). Analogical reasoning and conceptual complexity in cognitive development. *Human Development*, 35, 193-217.
- Halford, G. S. (1993). *Children's understanding: the development of mental models*. Hillsdale, N. J: Erlbaum.
- Halford, G. S., & Kelly, M. E. (1984). On the basis of early transitivity judgments. *Journal of Experimental Child Psychology*, 38, 42-63.
- Halford, G. S., Maybery, M., & Bain, J. D. (1986). Capacity limitations in children's reasoning: A dual-task approach. *Child Development*, 57, 616-627.
- Halford, G. S., Smith, S. B., Dickson, J. C., Maybery, M. T., Kelly, M., Bain, J. D., & Stewart, J. E. M. (1995). Modeling the development of reasoning strategies: The roles of analogy, knowledge, and capacity. In T. J. Simon & G. S. Halford (Eds.), *Developing cognitive competence: New approaches to process modeling* (pp. 77- 156). Hillsdale, NJ: Erlbaum.
- Halford, G. S., Wilson, W. H., & Phillips, S. (in press). Processing capacity defined by relational complexity: Implications for comparative, developmental, and cognitive psychology. *Behavioral Brain Sciences*.
- Holyoak, K. & Thagard, P. (1995). *Mental leaps*. Cambridge, MA: MIT Press.
- Ishihara, S. (1980). *Ishihara's Tests for Colour-Blindness*. Tokyo: Kanehara Shuppan Co., Ltd.
- Kallio, K. D. (1982). Developmental change on a five-term transitive inference. *Journal of Experimental Child Psychology*, 33, 142-164.
- Markovits, H., Dumas, C., & Malfait, N. (1995). Understanding transitivity of a spatial relationship: A developmental analysis. *Journal of Experimental Child Psychology*, 59, 124-141.
- Maybery, M. T., Bain, J. D., & Halford, G. S. (1986). Information processing demands of transitive inference. *Journal of Experimental Psychology: Learning, Memory & Cognition*, 12, 600-613.
- Pears, R., & Bryant, P. (1990). Transitive inferences by young children about spatial position. *British Journal of Psychology*, 81, 511-525.
- Piaget, J. (1957). *Logic and psychology*. New York: Basic Books.
- Rogers, C. M. & Berch, D. B. (1996). *Transitive inference in school-age children: The role of processing capacity*. Manuscript submitted for publication.
- Sternberg, R. J. (1980a). The development of linear syllogistic reasoning. *Journal of Experimental Child Psychology*, 29, 340-356.
- Sternberg, R.J. (1980b). Representation and process in linear syllogistic reasoning. *Journal of Experimental Psychology: General*, 109, 119-159.
- Thayer, E. S., & Collyer, C. E. (1978). The development of transitive inference: A review of recent approaches. *Psychological Bulletin*, 85(6), 1327-1343.
- Trabasso, T. (1975). Representation, memory, and reasoning: How do we make transitive inferences? In A.D. Pick (Ed.), *Minnesota symposia on child psychology* (Vol. 9, pp. 135-172). Minneapolis: University of Minnesota Press.
- Trabasso, T. (1977). The role of memory as a system in making transitive inferences. In R. V. Kail, Jr. & J. W. Hagen (Eds.) *Perspectives on the development of memory and cognition* (pp. 333-366). Hillsdale, NJ: Lawrence Erlbaum Associates.

- Trabasso, T., Riley, C. A., & Wilson, E. G. (1975). The representation of linear order and spatial strategies in reasoning: A developmental study. In R.J. Falmagne (Ed.), *Reasoning: Representation and process in children and adults* (pp. 201-229). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Youniss, J., & Murray, J. P. (1970). Transitive inference with nontransitive solutions controlled. *Developmental Psychology*, 2, 169-175.

APPENDIX

Analysis of Relational Information and Associated Baselines

Experiment 1

The guessing and one-relation baselines for the BD prediction questions in Experiment 1 correspond to Step 1 of item BD(C) with nonadjacent relation CE presented in Experiments 2 and 3. Table A-1 shows that the guessing and one-relation baselines coincide and correspond to probability correct of 0.50.

Experiments 2 and 3

The items used in Experiments 2 and 3 are shown in Table 3. Table A-1 summarizes the relational information of the items and the associated chance and one-relation baselines for the Step 1 and 2 of the items where nonadjacent relation CE is supplied in the premises. The correspondence between the items in Table A-1 and those for which nonadjacent relation BD is presented can be obtained from Table 3 in the text.

Chance baselines for Steps 1 and 2 of all items correspond to 0.50 and 0.33 respectively because there are two possible ways to arrange the blocks on Step 1, and there are three possible positions for the third block on Step 2. This yields an overall chance baseline of 0.415.

The analysis of items involving a single relation on each step (i.e., CD(B) and BC(D)) is relatively straight forward and will not be discussed further.

Step 1 of item BD(C) constitutes the basic transitivity problem in which two nonadjacent elements are ordered. Two relations (BC, CD) must be integrated on step 1. B and D each occur in two premises: B in AB and BC; D in CD and DE. Responses based on either BC or CD would be correct, responses based on AB or DE would be incorrect. Thus one-relation strategies yield a probability correct of 0.50.

Step 1 of items such as CE(D), can be completed by retrieving a single relation that corresponds to the nonadjacent relation presented, in this case, CE. On Step 2, D must be placed between the previously ordered pair. This requires consideration of two premise relations, CD and DE. Reliance on either CD (or DE) yields two possible orders CDE and CED (or CDE and DCE). A one-relation strategy yields a probability correct of 0.50 for Step 2 for this item type.

For items such as DE(C), the correct position of the third block is above the pair already ordered. Relations will be used if they include two elements that have to

be ordered. DE(C) can be processed in Step 1 by simply retrieving DE, the only relation in which both terms are mentioned. In Step 2, the problem is to place C with respect to DE. Two premise relations, CD and CE, entail two terms that have to be ordered, but it is not yet apparent which is appropriate. We therefore assume that they are equally likely to be chosen. If CD is chosen, the probability correct will be 1.00. If CE is chosen, the final ordering is ambiguous because either CDE or DCE would be consistent with the relation CE. The probability correct will be 0.50.

It is important to recognize that the participant does not know in advance which of CD and CE is appropriate. It may seem obvious to us that CD is the one to choose, but a moment's thought shows that we only recognize this because we already know the correct ordering. If you don't yet know this, then it is more reasonable to assume they are chosen with equal probability, given that there is no apparent bias. Note that if there were a bias towards CD this would make the task easier. We quickly determine the order as CDE, or we know it for alphabetic reasons, but the task presented to the child does not consist of alphabetic elements. The child's task is to determine the correct order of CDE. Until that is done, s/he has no way of knowing which of CD and CE is appropriate. It should be noted that items such as DE(C) differ from the construction items in Experiment 1 because nonadjacent relations were not included in premises in the earlier experiment. The one-relation baseline for Step 2 for items such as DE(C) corresponds to the average of 1.00 and 0.50 (i.e., 0.75 correct).

The one-relation baseline for Step 2 (i.e., items CE(D) and DE(C) combined) is 0.625 (average of 0.50 and 0.75). The overall one-relation baseline is based on two instances of Step 1 of BD(C) and one instance each of Step 2 of items CE(D) and DE(C). It corresponds to 0.5625 (average of 0.50 and 0.625).

Experiment 4

Chance. The probability of a correct response by chance is 0.5 for Step 1 and 0.33 for Step 2. The probability of Steps 1 and 2 being correct is 0.165.

One-Relation. The analysis corresponds to Step 1 of item BD(C) and Step 2 of items CE(D) in Table A-1. The probability of correct responses to both steps is therefore 0.25. This corresponds to a score of 2 out of 8.

Table A-1. Analysis of Relational Information and Baselines for Steps 1 and 2 of the Items in Table 3 with Nonadjacent Relation CE Supplied

Item	Step 1					Step 2				
	Relational information			Baselines		Relational information			Baselines	
	Relation retrieved	Implied order	Number of relations	Chance	One-relation	Relation retrieved	Implied order	Number of relations	Chance	One-relation
CD(B)	CD	C-D	1	0.50	0.50	BC	B-C-D	1	0.33	0.50
BC(D)	BC	B-C	1	0.50		CD	B-C-D	1	0.33	
BD(C)	BC, CD	B-D	2	0.50		None				
CE(D)	CE	C-E	1	0.50		CD & DE	C-D-E	2 ^a	0.33	
DE(C)	DE	D-E	1	0.50		CD	C-D-E or C-E-D	2 ^a	0.33	0.50
						DE	C-D-E or D-C-E			0.50
						CD & CE	C-D-E			1.00
						CD	C-D-E			0.50
						CE	C-D-E or D-C-E			

Note. ^a For reliable performance